

Workflow		Rank			HGSO	
All		All			All	
sequential	1	0.973	0.877	0.86	0.966	0.91
two_step	2	0.973	0.899	0.755	0.788	0.733
up_rf	3	0.972	0.849	0.652	0.757	0.362
smote_svm	4	0.969	0.811	0.638	0.752	0.524
hybrid_xgb	5	0.971	0.86	0.643	0.716	0.515
up_xgb	6	0.974	0.844	0.651	0.728	0.416
hybrid_svm	7	0.965	0.838	0.635	0.72	0.507
hybrid_rf	8	0.971	0.829	0.631	0.741	0.45
smote_xgb	9	0.972	0.833	0.628	0.71	0.579
smote_mr	10	0.956	0.827	0.629	0.691	0.418
up_svm	11	0.97	0.778	0.605	0.75	0.521
smote_rf	12	0.972	0.82	0.616	0.688	0.304
none_rf	13	0.968	0.848	0.568	0.723	0.4
hybrid_mr	14	0.952	0.821	0.613	0.682	0.412
up_mr	15	0.951	0.816	0.619	0.674	0.413
down_mr	16	0.913	0.817	0.47	0.638	0.302
down_svm	17	0.886	0.778	0.412	0.713	0.295
down_rf	18	0.901	0.799	0.419	0.589	0.262
down_xgb	19	0.886	0.771	0.42	0.608	0.194